STUDENT REPORT

DETAILS

Name

THANGADIPALLY ANVITHA

Roll Number

20R01A6753

EXPERIMENT

Title

WEEK 1

Description

1.Demonstration of preprocessing on dataset student.arff

Aim: This experiment illustrates some of the basic data preprocessing operations that can be performed using WEKA-Explorer. The sample dataset used for this example is the student data available in arff format.

Weka - Preprocessing the Data. The data that is collected from the field contains many unwanted things that leads to wrong analysis. For example, the data may contain null fields, it may contain columns that are irrelevant to the current analysis, and so on.

Step1: Loading the data. We can load the dataset into weka by clicking on open button in preprocessing interface and selecting the appropriate file.
Step2: Once the data is loaded, weka will recognize the attributes and during the scan of the data weka will compute some basic strategies on each attribute. The left panel in the above figure shows the list of recognized attributes while the top panel indicates the names of the base relation or table and the current working relation (which are same initially).
Step3: Clicking on an attribute in the left panel will show the basic statistics on the attributes for the categorical attributes the frequency of each attribute value is shown, while for continuous attributes we can obtain min, max, mean, standard deviation and deviation etc.,
Step4: The visualization in the right button panel in the form of cross-tabulation across two attributes.
Note:we can select another attribute using the dropdown list.
Step5:Selecting or filtering attributes Removing an attribute-When we need to remove an attribute,we can do this by using the attribute filters in weka.In the filter model panel, click on choose button, This will show a popup window with a list of available filters. Scroll down the list and select the "weka.filters.unsupervised.attribute.remove" filters.
Step 6:a)Next click the textbox immediately to the right of the choose button.In the resulting dialog box enter the index of the attribute to be filtered out.
b)Make sure that invert selection option is set to false.The click OK now in the filter box.you will see "Remove-R-7".
c)Click the apply button to apply filter to this data.This will remove the attribute and create new working relation.
d)Save the new working relation as an arff file by clicking save button on the top(button)panel.(student.arff)

Procedure:

Choose Tab->Weka->filters->Supervised->attribute->Discretize

Procedure and Overvation:

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Preprocessing

The process of transforming raw data into understandable format.

It can be classified into 4 types, they are data cleaning, data integration, data reduction, data transformation.

Remove attributes

a) Using 'remove' button

select an attribute

click on remove

b) using filter

click on choose

click on filter->unsupervised->attribute->remove

set attribute index

click ok

click apply

attribute will be removed

Handling of missing values

i)remove missing values

choose tab->Weka->filters->unsupervised->instances->remove with values.

set index values of attribute

set missing values to 'True' condition.

ii)replacing missing values

unsupervised->attribute->replace missing values

data type conversion

Discretization

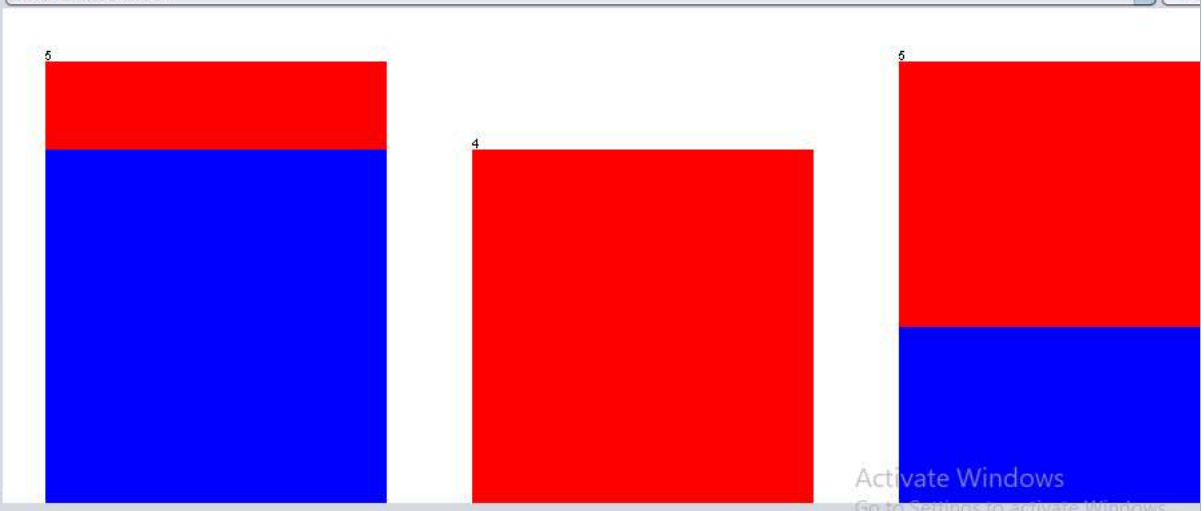
The process of converting a real value attribute into ordinal attribute or beans is called discretization.

Selected attribute

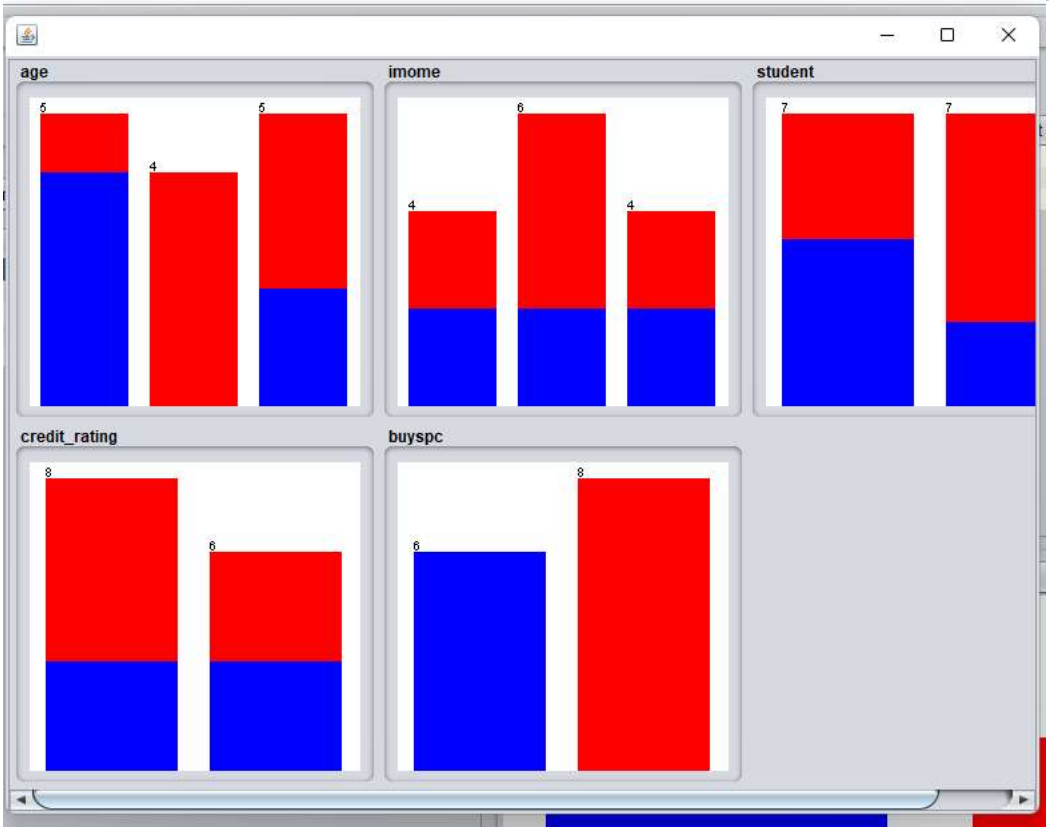
Name: age
Missing: 0 (0%)
Distinct: 3
Type: Nominal
Unique: 0 (0%)

No.	Label	Count	Weight
1	<30	5	5.0
2	30-40	4	4.0
3	>40	5	5.0

Class: buyspc (Nom)



Activate Windows
Go to Settings to activate Windows.



VIVA RESPONSES

Q) Identify the term used to define the task of inferring a model from labeled training data.

Supervised Learning

Q) Under which of the following do self-organizing maps lie.

Unsupervised Learning

Q) Is discrimination between spam and ham emails a classification task?

True

Q) Identify the incorrect option among the following which is not involved in data mining.

Data transformation

Q) Identify the correct options which are considered before investing in data mining.

All of the above